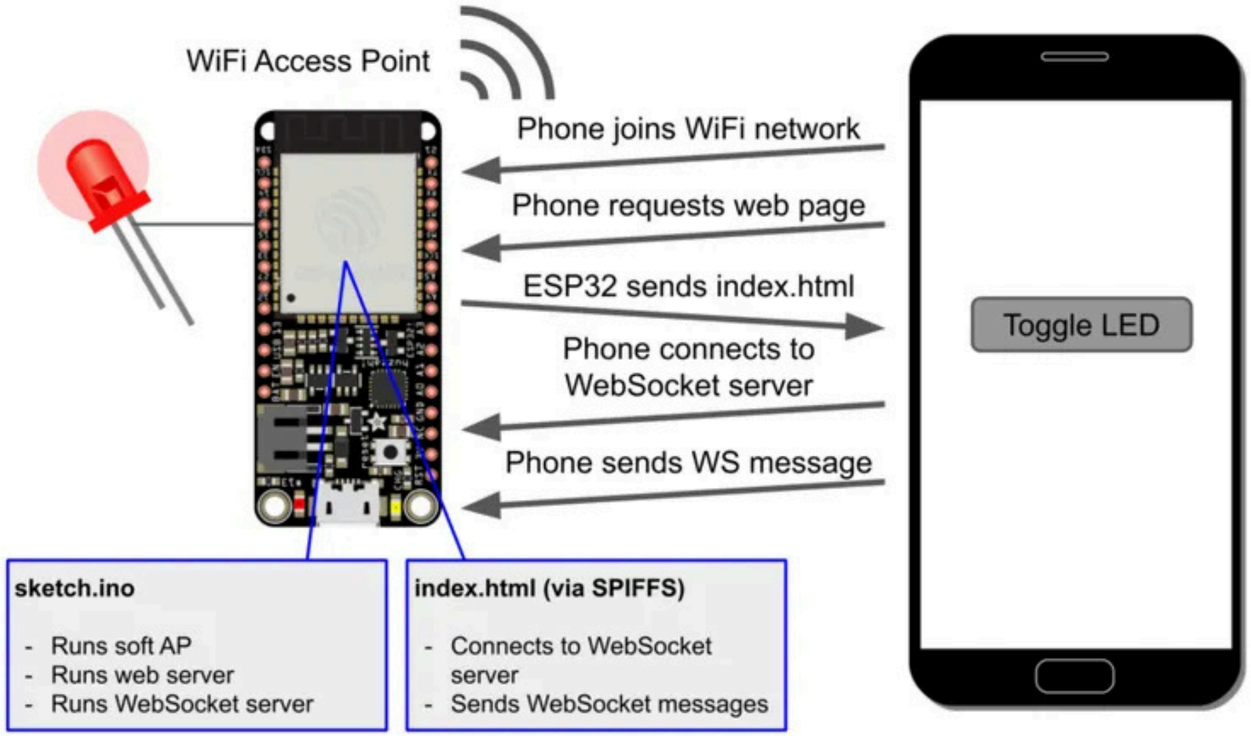
ESP32 WebSockets

Real-Time, Two-Way Communication between
ESP32 and Web Browser

What Are WebSockets?

A real-time communication protocol that keeps a connection open between the ESP32 and a web browser, allowing both sides to send data instantly.



Keeps the Connection Open

Unlike traditional HTTP polling, WebSockets maintain an open connection between the ESP32 and browser.

Real-Time Communication

The ESP32 can push data to the webpage immediately without waiting for the browser to request it.

Two-Way Communication

The browser can send messages to the ESP32, while the ESP32 can send messages back to the browser.

Runs Over Wi-Fi

The ESP32 connects to a local Wi-Fi network and hosts the WebSocket server.

Where Are WebSockets Used?

Anywhere real-time interaction is needed between an ESP32 and a web interface.

Real-Time Monitoring · IoT Dashboards · Remote Control · Sensor Data · Live Device Control · Web-Based Interfaces

Prerequisites & Hardware :

Hardware: 1× ESP32 development board.

USB data cable: Required for programming and Serial Monitor access. Make sure the cable supports data, not only charging.

Network: A local Wi-Fi network and router are required for the ESP32 to communicate with the browser.

Software: Arduino IDE installed on your computer

Step 1: Configure the Arduino IDE

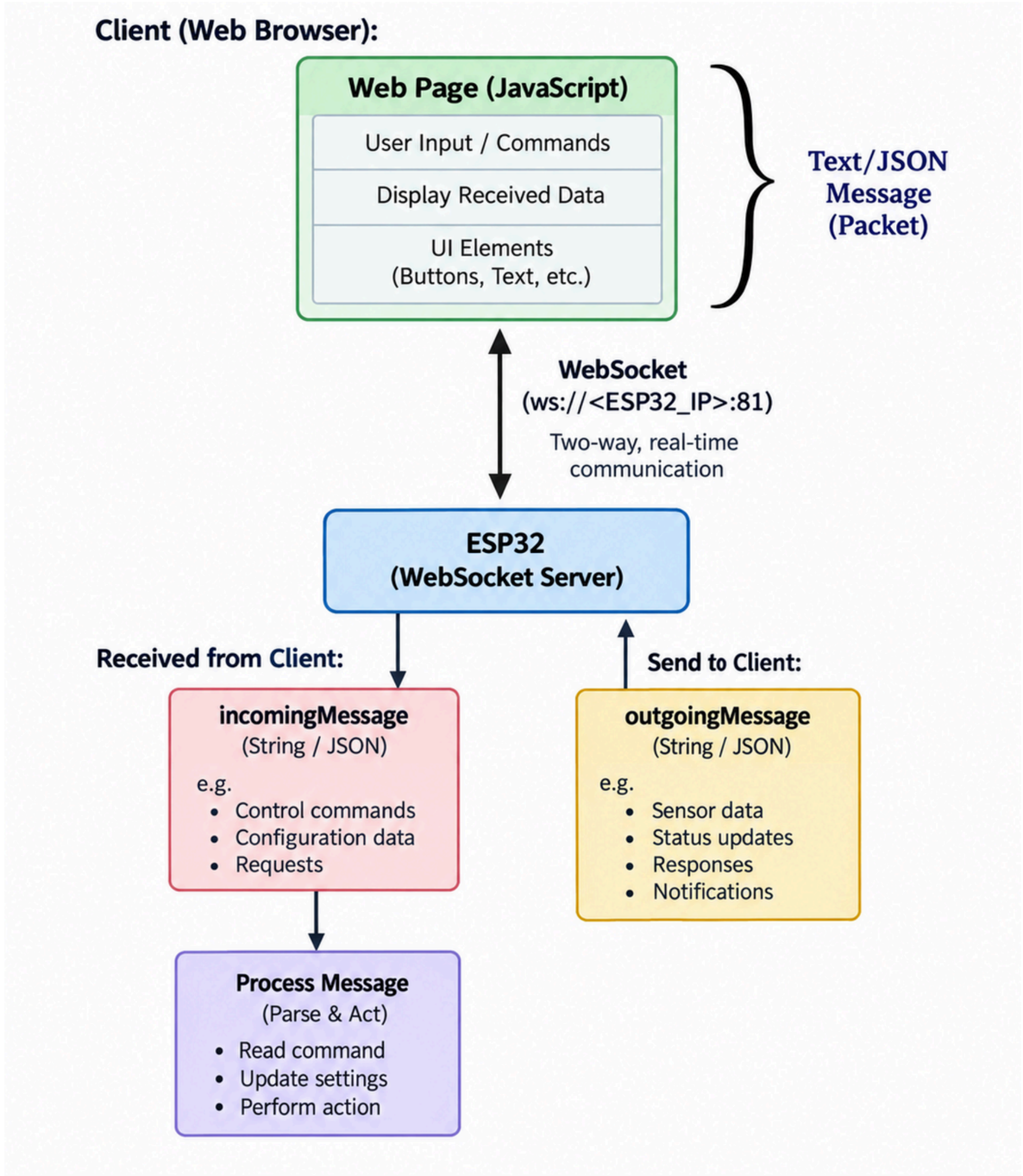
Step 2: Configure the ESP32 WebSocket Server

Step 3: Upload Code and Get the IP Address

Step 4: Create the Client Webpage

Step 5: Verify Communication

Understand the WebSocket Communication



That's WebSockets

The access the implementation guide
Scan the QR / click on the link in the description

Thank You